

LESSON 9-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

From Table to Graph

Maryland Standard 6.EE.C.9 • Learning target – I can turn a table of values into points and plot them on a graph.

REMEMBER

- Each row of a table becomes an **ordered pair** (x, y) .
- To plot (x, y) , start at $(0, 0)$. Move **right** x , then **up** y .
- Points from a constant rate line up in a **straight line**.

WORKED EXAMPLE

Movie tickets cost \$8 each. Write ordered pairs for 1, 2, and 3 tickets.

Step 1 Tickets is x . Cost is y . Multiply each x by 8.

Step 2 $8 \times 1 = 8$, $8 \times 2 = 16$, $8 \times 3 = 24$

Answer $(1, 8)$, $(2, 16)$, $(3, 24)$

WORD BANK

ordered pair – (x, y) : x first, then y

x-coordinate – how far right

y-coordinate – how far up

plot – mark a point on a graph

PRACTICE

show your work • label units

1. Write each table row as an ordered pair (x, y) .

Step 1 Write the x value first.

Step 2 Write the y value second, inside $()$.

a. $x = 4, y = 12$ _____

b. $x = 0, y = 0$ _____

c. $x = 6, y = 30$ _____

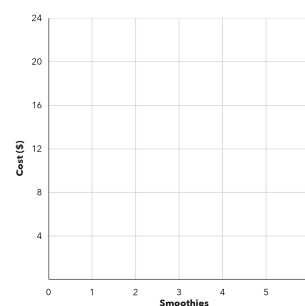
d. $x = 9, y = 2$ _____

2. **Smoothies.** Smoothies cost \$4 each. Complete the table. Then plot the points on the grid and connect them.

Step 1 Multiply each number of smoothies by \$4.

Step 2 Plot each pair: go right x , then up y .

Smoothies (x)	1	2	3	5
Cost in \$ (y)				

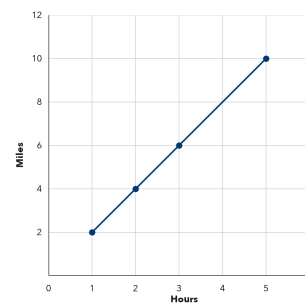


3. **Hiking.** The graph shows how far Kofi hikes.

a. Write the ordered pair for 5 hours. _____

b. What does $(3, 6)$ mean? _____

c. Miles per hour: _____



- 4. Bike Rental.** A bike costs **\$6** for each hour. Write the ordered pairs (hours, cost) for 1, 3, and 4 hours. Then tell what (0, 0) means.

Step 1 Multiply each number of hours by 6.

Step 2 Write (hours, cost).

Ordered pairs _____

- 5. Find the Error.** Teo's table shows x (hours): 2, 4 and y (dollars): 10, 20. He plotted (10, 2) and (20, 4). Explain his mistake. Which points should he plot?

Frame Teo's mistake is _____. He should plot _____ and _____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Show someone at home how to plot (3, 5): right 3, then up 5.

LESSON 9-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Analyzing a Graph**

Maryland Standard 6.EE.C.9 • Learning target – I can read a relationship off a graph and describe the rate it shows.

REMEMBER

- For a line through (0, 0), the **unit rate** is $y \div x$. The point (3, 12) means 3 of x go with 12 of y .
- With the same scales, a **steeper** line has a greater rate.

WORKED EXAMPLE

A line through (0, 0) passes through (2, 10). Find the unit rate and y when $x = 7$.

Step 1 Unit rate: $10 \div 2 = 5$

Step 2 $y = 5 \times 7 = 35$

Answer 5 per unit; $y = 35$

WORD BANK

unit rate – the amount for 1 of x

steeper – rises faster

origin – the point (0, 0)

proportional – a straight line through (0, 0)

PRACTICE*show your work • label units*

- 1. Filling a Pool.** The graph shows the water in a pool.

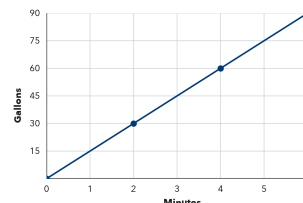
Step 1 Find (4, 60) on the graph. Read the labels on both axes.

Step 2 For 1 minute, divide the gallons by the minutes.

a. What does (4, 60) mean? _____

b. Gallons per minute: _____

c. Gallons after 9 minutes: _____



- 2.** Each line passes through (0, 0) and the given point. Find the unit rate.

Step 1 Divide y by x .

a. (3, 21) _____

b. (5, 40) _____

c. (8, 96) _____

- 3. Reading.** A line through (0, 0) shows that Ana reads 18 pages in 3 hours. Find her unit rate. How many pages does she read in 7 hours?

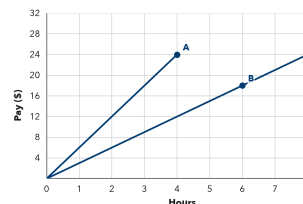
Step 1 Divide the pages by the hours to find pages in 1 hour.

Step 2 Multiply that by 7.

Answer _____

- 4. Two Jobs.** The graph shows pay for two jobs. Find each pay rate. Which line is steeper, and what does it mean?

Answer _____



5. **Savings.** Jada's savings graph passes through $(0, 10)$, $(1, 15)$, and $(2, 20)$. She says it is proportional. Is she right? What does $(0, 10)$ mean?

Frame Jada is ____ because a proportional graph _____. $(0, 10)$ means _____.

TEST PRACTICE

choose the best answer

6. A line through the origin passes through $(5, 35)$. What is the unit rate?
A 5 **B** 30 **C** 7 **D** 175
7. Line P passes through $(1, 4)$. Line Q passes through $(1, 9)$. Both start at $(0, 0)$. Which is true?
A P is steeper. **B** Q is steeper; greater rate.
C They have the same rate. **D** Q is steeper; smaller rate.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____